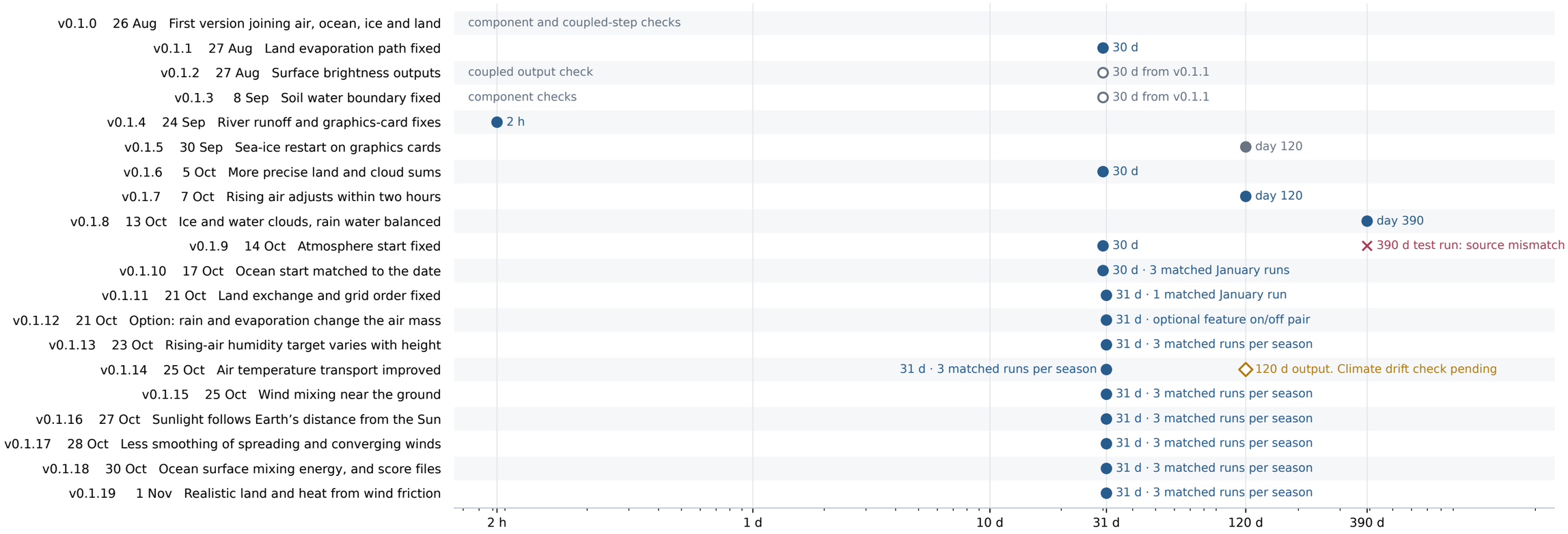
ReadyESM development scorecard

Evidence inventory: 1 November 2026, after six matched months

v0.1.19 — release record with disclosed test outcomes

A. Release tests: simulated time reached

Logarithmic duration axis · starting from observations does not establish long-term climate skill



B. Exploratory model changes

Current work and test results

		Furthest test completed	Checks completed for each item						
		Idea	Code	Pilot	Score	Check	In use	Off	Resume	Balance	Limits	Repeat	Climate
BS01 Rising-air plumes Core kept. The tested plume version failed the climate test.	supported		●					○	—	✓	○	○	✕
BS02 Ice and water clouds Moving and restarting work. The tested cloud rule failed.	supported				●			✓	✓	✓	✓	~	✕
BS03 Finer layers for sunlight and heat New model top and layer options are being tested.	active			●				✓	✓	~	~	○	○
BS04 Balanced start Fixes to the atmosphere and ocean start are in use.	accepted						●	✓	✓	~	✓	✓	✓
BS05 Systematic tuning Only one-column checks of which settings can be told apart.	supported		●					—	—	○	○	○	○
BS06 Ice of several thicknesses The tested ice pressure rule failed the pile-up test.	supported				●			○	✓	~	✓	✓	✕
BS07 Coastal ice and ice fixed to the shore The coast connection part is kept. Simple fixes failed.	supported				●			○	—	✓	✓	~	✕
BS08 Water stored in river basins The water store works. Heat carried by runoff stays off.	supported			●				○	✓	✓	~	○	✕
BS09 Tools for linked test runs The tools are in use. Model changes still need their own tests.	accepted						■	—	—	—	—	—	—
BS10 Optional carbon cycle Optional carbon is in use after four days on the full grid.	accepted						●	✓	✓	✓	✓	○	○

C. Clouds and near-surface air

Current work and test results

		Furthest test completed	Checks completed for each item						
		Idea	Code	Pilot	Score	Check	In use	Off	Resume	Balance	Limits	Repeat	Climate
C01 Rising air starting near the ground The tested start depths failed the joint 10-day test.	closed				●			✓	—	✓	✓	○	✕
C02 Shallow rising air with a depth limit The tested depths failed the joint 10-day test.	closed				●			✓	—	✓	✓	○	✕
C03 Shallow rising air driven by the ground No tested setting passes the joint test.	closed				●			✓	○	~	✓	○	✕
C04 Raised and slanted rising air Slanted rising air appears in less than 1% of inspected cases.	reserve		●					—	—	—	—	—	—
C05 Deep rising air with memory The memory test was too weak to see past day-to-day noise.	supported				●			○	○	~	✓	~	~
C06 Falling air and cool air pools The tested cool-air-pool setting fails the second-season case.	supported				●			○	○	○	✓	○	✕
C07 Wind carried by rising air The parts work and a one-day trial ran.	active			●				~	○	✓	✓	○	○
C08 Many rising-air plume sizes The January version is parked. The two-season test was undecided.	closed				●			✓	~	✓	✓	○	✕
C09 Random rising-air plumes The eight-run version was closed: its effect was too small to measure.	closed				●			✓	~	✓	✓	~	✕
C10 Mixing near the ground Only wind mixing was accepted in v0.1.15.	accepted						●	✓	✓	~	✓	✓	~

● tested for published release ○ earlier or unreleased test ◇ run finished, review pending ✕ source mismatch, excluded

● pass ■ fail ◆ partial / limited ○ not established — not applicable

Each row shows the furthest test completed. Items marked in use include optional tools. Climate skill is measured separately.

Off: works when switched off · Resume: continues from a saved state · Balance: energy and water accounted for · Limits: values stay physical · Repeat: matched runs agree · Climate: meets target

v0.1.12 tested an optional feature for 31 days but left it off in standard runs. v0.1.13 checked a short restart. v0.1.14 and v0.1.15 allowed documented exceptional values. Long drift remains open.

Qualified v0.1.19 release. The frozen score REJECTS it: energy and January sea-ice checks fail. Long-term climate skill is unproved.